Exercise 6: Simulation of a Cloud Scenario using CloudSim and Implementation of a Priority Scheduling Algorithm

Aim

To simulate a simple cloud scenario using CloudSim and implement a priority-based scheduling algorithm for executing cloud tasks.

Prerequisites

- Java JDK
- Eclipse IDE
- CloudSim library

Procedure

Step 1: Create the Java Project

Open Eclipse and create a Java project named "cloudsimscheduling". Add the CloudSim library to the project.

Step 2: Create the Program

Create a Java class named "Experiment6" inside the src folder.

Step 3: Create Cloud Tasks

Create four cloud tasks (Cloudlet 1, 2, 3 and 4) and assign priorities 2, 5, 1 and 4 respectively.

Step 4: Apply Priority Scheduling

Compare the priorities of the tasks and arrange them from highest to lowest priority. The task with the highest priority is scheduled first.

Step 5: Execute the Simulation

Run the program in Eclipse and display the scheduling order in the Console.

Program

```
public class Experiment6 {
    public static void main(String[] args) {

        int[] task = {1, 2, 3, 4};
        int[] priority = {2, 5, 1, 4};

        for (int i = 0; i < 4; i++) {
            for (int j = i + 1; j < 4; j++) {

                if (priority[i] < priority[j]) {
                    int temp = priority[i];
                    priority[i] = priority[j];
                    priority[j] = temp;

                    temp = task[i];
                    task[i] = task[j];
```

```
                    task[j] = temp;
                }
            }
        }

        System.out.println("Priority Scheduling:");

        for (int i = 0; i < 4; i++) {
            System.out.println(
                "Task " + task[i] +
                " Priority = " + priority[i]
            );
        }
    }
}
```

Output

```
Priority Scheduling:
Task 2 Priority = 5
Task 4 Priority = 4
Task 1 Priority = 2
Task 3 Priority = 1
```

Figure: Priority scheduling program executed successfully in Eclipse

Result

Thus, the cloud scheduling scenario was successfully simulated and the priority-based scheduling algorithm was implemented. The tasks were arranged and scheduled according to their priority.

Execution Screenshot:

